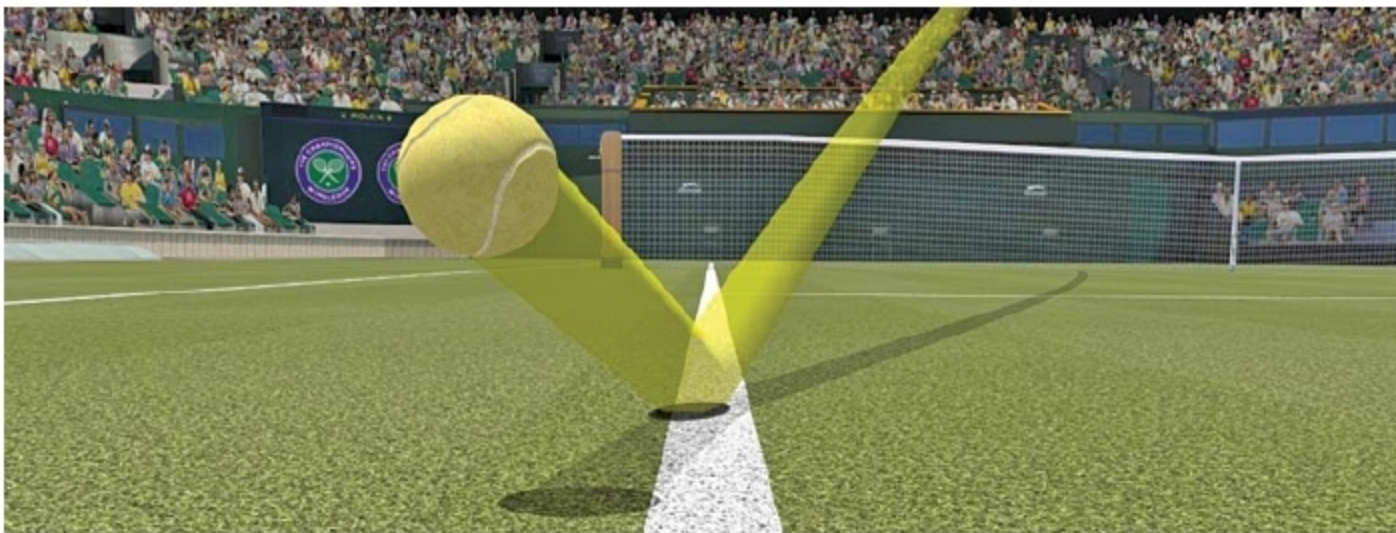


TECHNOLOGIES THAT HAVE TRANSFORMED **THE SPORTS** EXPERIENCE



In-shot technology being used in tennis (Credit: www.hawkeyeinnovations.com)

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You do not need to watch too many games to know that referees have limited visibility and they too can make mistakes. Technology can change this by helping them make better and more accurate decisions. There are many technologies that can assist a referee on the ground (central referee) or through the third empire who is looking at the game from outside the ground (video-assisted). If you do not think these technologies are

necessary, just imagine for a while how hard it is to keep an eye on all players on the ground at one time while closely keeping watch on the ball.

Given below are a few technologies that have started to impact and transform the field of sports, although on a restricted basis so that no harm comes to the spirit of the game.

Video replay system

Virtual reality (VR) and 360-degree cameras give referees an unprecedented view of the action. A video replay system (VRS) is used in many sports including basketball, football, cricket and rugby. Basketball referees use it to make sure players are shooting within the time allotted by the shot clock. They use replay vision to review last-touch

decisions in the final two minutes of the game.

In rugby, VRS with synchronised camera views is used to improve decision-making. It is also used by the medical staff to identify possible concussion instances.

In football, the technology is used to decide whether a ball has passed over the goal line, and to clarify penalty decisions. In football, there are four types of calls that can be reviewed: goals, penalty decisions, red card decisions and mistaken identity in awarding a card.

In many games like cricket and football, the video-assisted referee reviews video replays of the event, and informs the central referee via